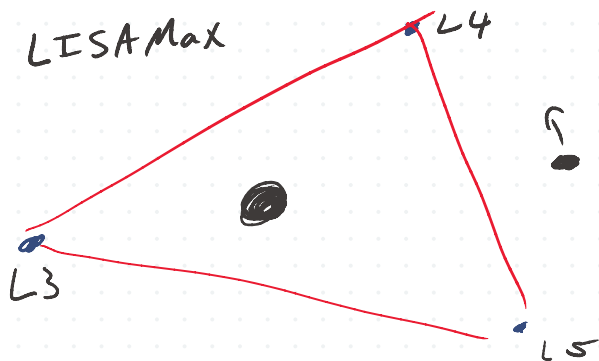
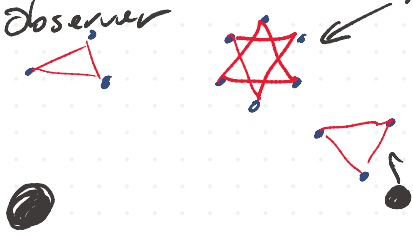


LISA Max



Sky locations are very large.

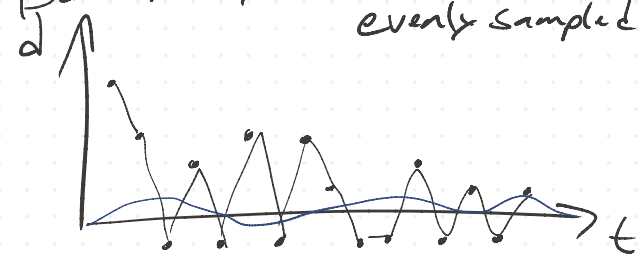
Big Bang Observer



Measure polarization of GWs instantly.

Data Analysis

evenly sampled data



$$d(t) = n(t) + s(t)$$

Normally  $|s(t)|^2 \ll |n(t)|^2$ .

How do we measure if  $d = n$  null hypothesis

or  $d = n + s$  ?  
Hypothesis

Can we reject the null hypothesis?

One sample!

$$\begin{array}{|c|} \bullet \\ \hline \end{array} = \begin{array}{|c|} \bullet \\ \hline \end{array}$$
$$d_i = n_i + s_i$$

$s_i$  is deterministic

$n_i$  is stochastic

$$\langle n_i \rangle = \mu_i$$
$$= \int_{-\infty}^{+\infty} n_i dt$$

Normally this  $= 0$ .

Basic idea:

Test  $d(t) - s(t)$ , see  
if it looks like noise or not.  
If not,  $s(t)$  is wrong!

---

Noise has a probability distribution:

$$n \sim p(x)$$

Moments of probability distributions:

- Raw moments

$$\langle x^n \rangle = \int_{-\infty}^{+\infty} x^n p(x) dx$$

$$\text{Mean } \langle x \rangle = \mu$$

- Central moments

$$\mu_m = \langle (x - \mu)^m \rangle = \int_{-\infty}^{+\infty} (x - \mu)^m p(x) dx$$

$$\mu_2 = \langle (x - \mu)^2 \rangle = \sigma^2$$

$$\text{Variance} = \sigma^2 \Rightarrow \sigma = \text{"standard deviation"}$$

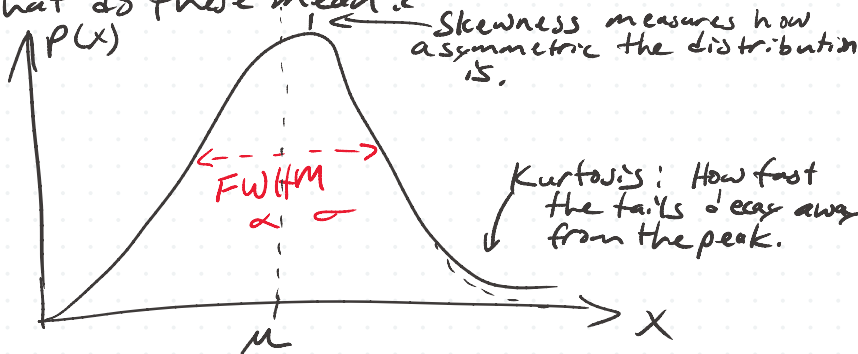
- Standardized moments

$$\tilde{\mu}_m = \left\langle \left( \frac{x - \mu}{\sigma} \right)^m \right\rangle = \int_{-\infty}^{+\infty} \left( \frac{x - \mu}{\sigma} \right)^m p(x) dx$$

$$m=3, \tilde{\mu}_3 = \left\langle \left( \frac{x-\mu}{\sigma} \right)^3 \right\rangle \quad \text{skewness}$$

$$m=4, \tilde{\mu}_4 = \left\langle \left( \frac{x-\mu}{\sigma} \right)^4 \right\rangle \quad \text{kurtosis}$$

What do these mean?



Gaussian distribution: skewness = 0  
kurtosis = 0.

In real life, we usually only need 3 distributions:

- Switching things on and off: **Binomial**
- Counting things: **Poisson** →  $\sigma^2 = \mu$
- Everything else (except for weird scenarios): **Gaussian**  
(Watching things wiggle)

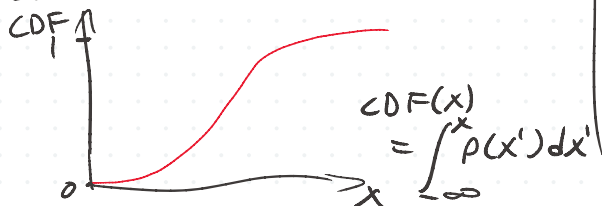
Central limit theorem (generalized):

$$P_1(x) + P_2(x) + \dots \rightarrow \text{Gaussian}$$

(up to same technical conditions) on  $P_i(x)$ ,  $\sum P_i \rightarrow \text{Gaussian}$

⇒ If a lot of unrelated effects contribute to a distribution ⇒ Gaussian.

Cumulative distribution function



Basically, not too related to each other, and with no one  $P_i$  dominating over the others.

For GW detectors, things are wiggling  $\Rightarrow$  Gaussian!

$$N(\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

\mathcal{N}(\mu, \sigma^2)

For us, noise oscillates about an arbitrary value.  
we'll set it to 0.

$$n \sim N(0, \sigma^2)$$

Say we measure a sample  $n_i$ . What is the probability?

$$p(n_i) = (2\pi\sigma^2)^{-1/2} e^{-\frac{n_i^2}{2\sigma^2}}$$

$\Rightarrow$  What is the probability of measuring  $d$  if it contains  $n + s$ ?

$$r_i = d_i - s_i \Rightarrow p(d_i | s_i) = (2\pi\sigma^2)^{-1/2} e^{-\frac{r_i r_i}{2\sigma^2}}$$

$\sigma^2$ : variance of noise.

$p(A|B) = p(A)$  if  $B$  is true.

Likelihood